Please write clearly in block capitals.

Centre number Candidate number

Surname ______________________________

Forename(s) ______________________________

Candidate signature ______________________________

A-level

ECONOMICS

Paper 3 Economic principles and issues

Predicted Paper **Time allowed: 2 hours**

Materials

For this paper you must have:

- an AQA 16-page answer book
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Write the information required on the front of your answer book. The Paper Reference is 7136/3.
- Answer **ALL** questions.
- In Section A, for each multiple-choice question, select one answer from A, B, C or D.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 80.
- There are 30 marks for Section A and 50 marks for Section B.
- You will be marked on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

Advice

- You are advised to spend 40 minutes on Section A and 80 minutes on Section B.

Predicted Paper — For revision purposes only [Set C — Advanced]

Section A

Answer all questions. Each question is worth 1 mark.

For each question, select the single best option: A, B, C or D.

Question 01 **[1 mark]**

In two-sided digital platform markets, a profit-maximising platform is most likely to set:

A Prices that always exceed marginal cost on both sides of the market.
B Prices below marginal cost on one side and above marginal cost on the other.
C Uniform zero prices on both sides of the market.
D Prices that equal the monopoly price on each side.

Question 02 **[1 mark]**

Hysteresis in the labour market refers to the tendency for:

A A temporary shock to aggregate demand to have permanent effects on the natural rate of unemployment.
B Workers to move voluntarily between jobs in search of higher wages.
C Long-run inflation to drift upward over time.
D Productivity growth to accelerate automatically after a recession.

Question 03 **[1 mark]**

A first-degree price discriminator captures:

A No consumer surplus; consumer welfare is unchanged.
B Some consumer surplus; the deadweight loss relative to perfect competition is reduced.
C All consumer surplus; output equals the perfectly competitive level and there is no deadweight loss.
D All consumer surplus and all producer surplus.

Question 04 **[1 mark]**

The Coase theorem implies that externalities can be resolved efficiently by private bargaining when:

A Property rights are undefined and transaction costs are low.
B Property rights are well-defined and transaction costs are low.
C The government imposes a Pigouvian tax.
D Firms are monopolists with full market power.

Question 05 **[1 mark]**

In a simple Solow growth model with diminishing returns to capital, long-run per-capita output growth is driven primarily by:

A Capital accumulation alone.
B Labour force growth alone.
C Exogenous technological progress.
D Government consumption.

Question 06 **[1 mark]**

A lump-sum tax is considered by most economists to be the least distortionary form of taxation because:

- **A** It raises the most revenue.
- **B** It falls only on high-income households.
- **C** It does not depend on behaviour, so it does not distort economic decisions at the margin.
- **D** It is politically popular.

Question 07 **[1 mark]**

Under floating exchange rates, a capital-importing country will typically observe:

- **A** A current account surplus matched by a capital account deficit.
- **B** A current account deficit matched by a capital account surplus.
- **C** Both a current account surplus and a capital account surplus.
- **D** A zero balance of payments on all accounts.

Question 08 **[1 mark]**

Adverse selection in insurance markets arises primarily because:

- **A** Insurers can observe policyholders' actions after the contract is signed.
- **B** Policyholders with the highest risk have the greatest incentive to buy insurance when insurers cannot distinguish risk levels.
- **C** Governments regulate premiums.
- **D** Competition among insurers drives premiums to zero.

Question 09 **[1 mark]**

The natural rate hypothesis implies that:

- **A** There is a stable long-run trade-off between unemployment and inflation.
- **B** The long-run Phillips curve is horizontal.
- **C** The long-run Phillips curve is vertical at the natural rate of unemployment.
- **D** Monetary policy can permanently reduce unemployment by accepting higher inflation.

Question 10 **[1 mark]**

If a central bank follows a Taylor rule and inflation is 1 percentage point above target while output is at potential, the rule prescribes that the central bank should:

- **A** Leave the policy rate unchanged.
- **B** Cut the policy rate to stimulate demand.
- **C** Raise the policy rate by more than one percentage point.
- **D** Raise the policy rate by less than one percentage point.

Question 11 **[1 mark]**

The table below shows the short-run total costs of a firm facing a perfectly elastic demand at P = £25.

Table 1

Output (units)	Total Variable Cost (£)	Total Fixed Cost (£)
0	0	60
10	100	60
20	180	60
30	270	60
40	380	60
50	520	60

Based on Table 1, the firm's profit-maximising output is:

A 20 units.
B 30 units.
C 40 units.
D 50 units.

Question 12 **[1 mark]**

The table below shows a game in normal form between two oligopolists, Firm 1 and Firm 2, choosing between High and Low prices. Payoffs are (Firm 1, Firm 2) in £m.

Table 2 — Payoff matrix

	Firm 2: High	Firm 2: Low
Firm 1: High	(12, 12)	(4, 14)
Firm 1: Low	(14, 4)	(6, 6)

Based on Table 2, the Nash equilibrium of this game is:

A Both firms choose High — cooperative outcome.
B Both firms choose Low — classic prisoners' dilemma outcome.
C Firm 1 chooses High, Firm 2 chooses Low.
D There is no Nash equilibrium in pure strategies.

Question 13 **[1 mark]**

The table below shows CPI and wage data for an economy.

Table 3

Year	CPI (2020 = 100)	Avg nominal wage (£ per hour)
2020	100.0	15.00
2021	102.5	15.50
2022	111.7	16.30
2023	119.8	17.40
2024	123.6	18.50

Based on Table 3, between 2020 and 2024, the real hourly wage has changed by approximately:

A –0.3%.
B +0.4%.
C +23.3%.
D +23.6%.

Question 14 **[1 mark]**

The table below shows the sterling effective exchange rate and relative productivity of two economies.

Table 4

Year	UK sterling ERI	UK real unit labour costs (index 2015 = 100)
2015	100.0	100.0
2019	80.4	97.2
2022	79.1	104.1
2024	83.6	101.8

Based on Table 4, between 2015 and 2024 the UK's international price competitiveness has:

A Improved significantly, because sterling has depreciated.
B Improved, because real unit labour costs have fallen.
C Worsened overall, because the depreciation has been partly offset by rising real unit labour costs.
D Remained unchanged.

Question 15 **[1 mark]**

The table below shows the hours worked and output of an AI-assisted and a non-AI-assisted worker in the same firm.

Table 5

Worker type	Hours worked per week	Output per week (units)
Non-AI-assisted	40	80
AI-assisted	40	160

Based on Table 5, assuming wage rates are unchanged, the firm's demand for non-AI-assisted labour is most likely to:

A Rise, because output has increased.

B Fall, because AI-assisted workers are now more cost-effective.

C Remain unchanged.

D Fall only if the AI-assisted worker costs more than twice the non-AI-assisted worker.

Question 16 **[1 mark]**

The table below shows the marginal external cost (MEC) of carbon emissions from four industries.

Table 6

Industry	Marginal external cost (£ per tonne CO■)	Current carbon price (£ per tonne CO■)
Power generation	75	45
Heavy industry	80	45
Aviation	90	25
Agriculture	60	0

Based on Table 6, which industry is the furthest from having its negative externality correctly priced?

A Power generation.
B Heavy industry.
C Aviation.
D Agriculture.

Question 17 **[1 mark]**

The table below shows hypothetical Gini coefficients before and after tax-and-transfer policies.

Table 7

Country	Gini before (market)	Gini after (disposable)
A	0.51	0.36
B	0.48	0.40
C	0.53	0.27
D	0.44	0.38

Based on Table 7, which country has the strongest redistributive tax-and-transfer system?

A Country A.
B Country B.
C Country C.
D Country D.

Question 18 **[1 mark]**

The table below shows the world price, domestic supply and domestic demand for a good.

Table 8

Price (£)	Domestic supply	Domestic demand
10 (world price)	20	100
13 (with tariff)	35	85
15	45	75

Based on Table 8, the government imposes a £3 per-unit tariff, raising the domestic price to £13. The tariff revenue collected is:

A £60.
B £150.
C £80 × 3 = £240.
D (85 – 35) × £3 = £150.

Question 19 **[1 mark]**

The table below shows the share of UK wealth owned by each decile.

Table 9

Decile	**% of total wealth**
Bottom 10%	–0.2
Deciles 2–5 (bottom 50%)	9.0
Deciles 6–9 (middle 40%)	48.0
Top 10%	43.0

Based on Table 9, which one of the following statements is correct?

A Wealth inequality is lower than in a perfectly equal distribution.

B The top decile owns more wealth than the bottom 50% combined.

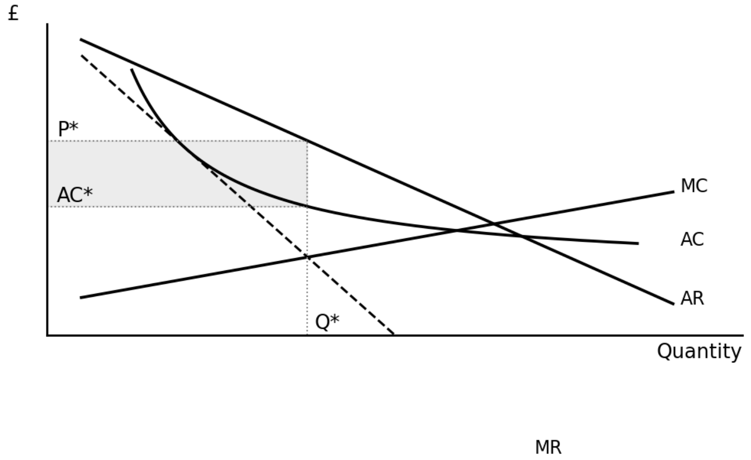
C The bottom 10% owns the most wealth.

D There is no wealth inequality in the UK.

Question 20 **[1 mark]**

Figure 1 shows the cost and revenue curves for a monopolist.

Figure 1 — Monopoly profit maximisation



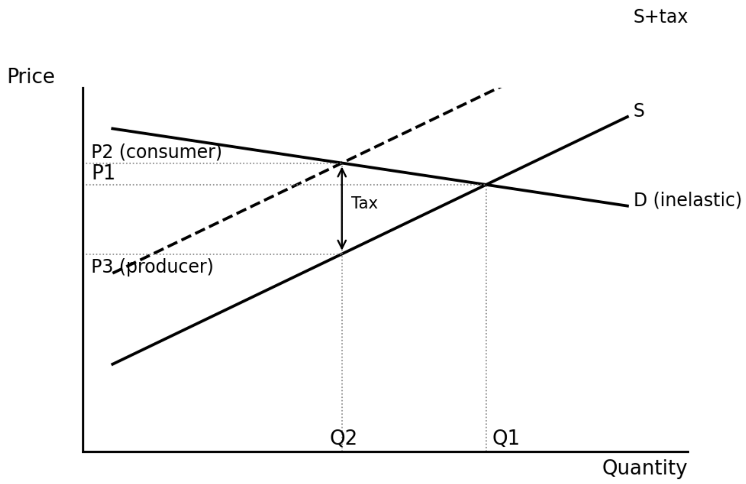
Based on Figure 1, compared with a perfectly competitive market with identical cost curves, the monopoly outcome involves:

A Higher output and lower price.
B Lower output, higher price, and a deadweight welfare loss.
C The same output and price.
D Higher output and higher price.

Question 21 **[1 mark]**

Figure 2 shows the incidence of an indirect tax on a good.

Figure 2 — Tax incidence



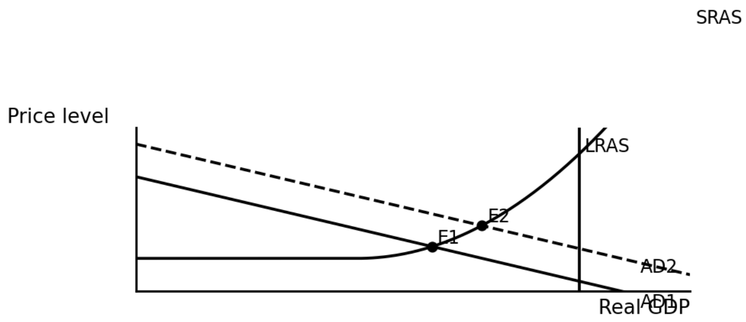
Based on Figure 2, the deadweight welfare loss of the tax is represented by:

A The total tax revenue collected.
B The reduction in consumer surplus.
C The triangle formed by the reduction in quantity from Q1 to Q2 between supply and demand.
D The gain in producer surplus.

Question 22 **[1 mark]**

Figure 3 shows an AD shift in an economy with spare capacity.

Figure 3 — AD shift near LRAS vs far from LRAS



Based on Figure 3, the elasticity of SRAS implies that the inflation cost of demand stimulus:

A Is the same wherever the economy is operating.

B Is lower the further the economy is from LRAS and higher the closer it gets.

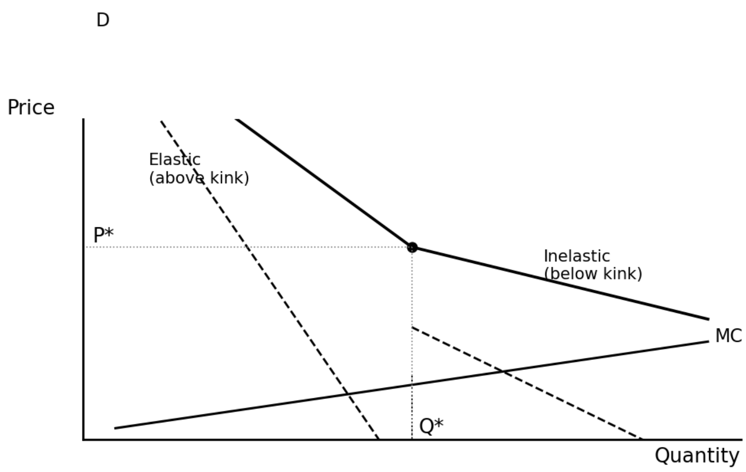
C Is always zero.

D Is always equal to the nominal wage growth rate.

Question 23 **[1 mark]**

Figure 4 shows the kinked demand curve model of oligopoly.

Figure 4 — Kinked demand curve



Based on Figure 4, the kinked demand curve model is limited as a theory of oligopoly because it:

A Explains the initial setting of the kinked price.

B Does not explain how the initial price is set — only why it is rigid afterward.

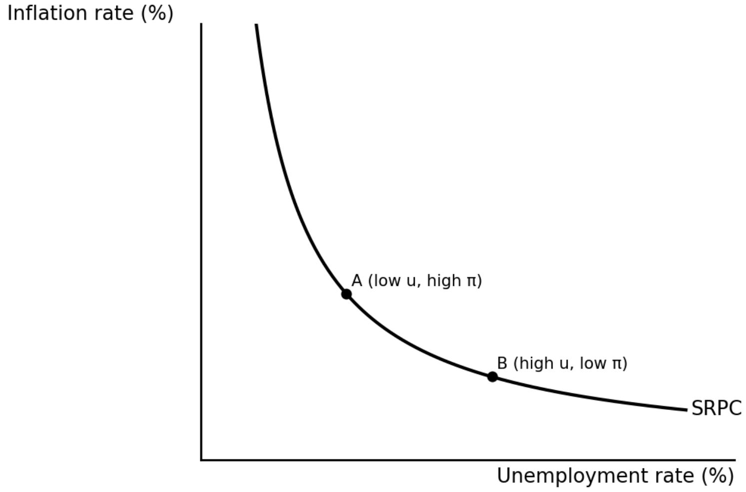
C Predicts aggressive price wars in all circumstances.

D Is only applicable to monopolistic competition.

Question 24 **[1 mark]**

Figure 5 shows a short-run Phillips curve.

Figure 5 — Short-run Phillips curve



Based on Figure 5, an adverse supply shock (such as a sharp rise in energy prices) is most likely to cause:

A A movement along the existing short-run Phillips curve.

B The short-run Phillips curve to shift outward (higher inflation at every unemployment rate).

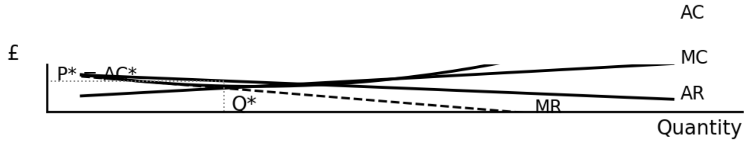
C The short-run Phillips curve to shift inward.

D The long-run Phillips curve to shift to the left.

Question 25 **[1 mark]**

Figure 6 shows the long-run equilibrium in monopolistic competition.

Figure 6 — Monopolistic competition: long-run equilibrium



Based on Figure 6, the efficiency loss in monopolistic competition compared with perfect competition is best described as:

A Zero, because only normal profits are earned.

B Excess capacity: firms operate at output below the minimum point of the AC curve.

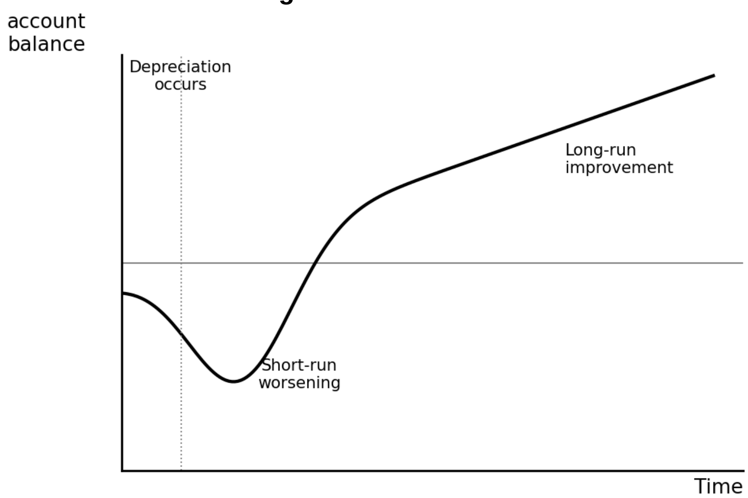
C Supernormal profit in the long run.

D Allocative efficiency at P = MC.

Question 26 **[1 mark]**

Figure 7 shows the J-curve effect.

Figure 7 — J-curve



Based on Figure 7, the short-run deterioration of the current account after depreciation is primarily explained by:

A Elasticities of demand for exports and imports being low in the short run.

B Elasticities of demand being high in the short run.

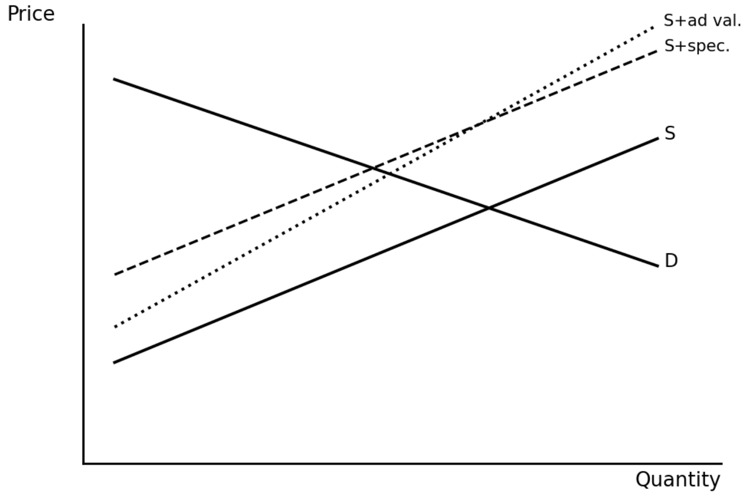
C An immediate fall in the foreign-currency value of imports.

D Tariffs imposed by trading partners.

Question 27 **[1 mark]**

Figure 8 compares specific and ad valorem taxes.

Figure 8 — Specific vs ad valorem tax



Based on Figure 8, if the government wishes the tax to have a progressively larger absolute impact on higher-priced goods, it should use:

A A specific tax.

B An ad valorem tax.

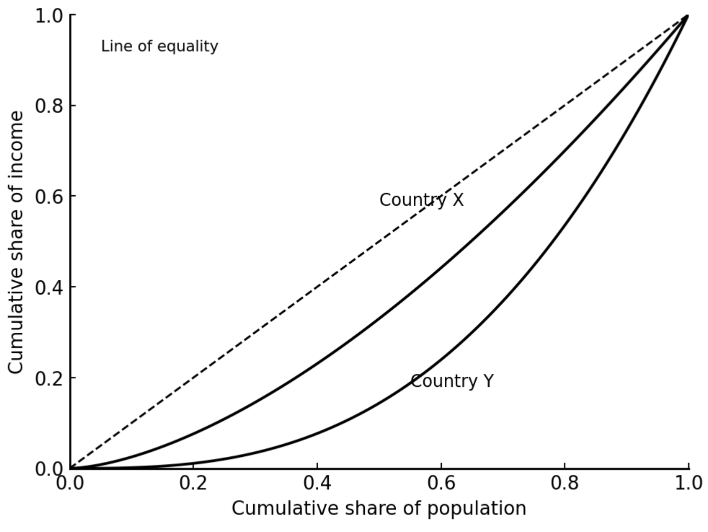
C A lump-sum tax.

D Either — the choice has no effect.

Question 28 **[1 mark]**

Figure 9 shows Lorenz curves.

Figure 9 — Lorenz curves for countries X and Y



Based on Figure 9, the Gini coefficient is calculated as:

A The area between the Lorenz curve and the line of equality, expressed as a proportion of the total area under the line of equality.

B The area below the Lorenz curve, expressed as a proportion of the area under the diagonal.

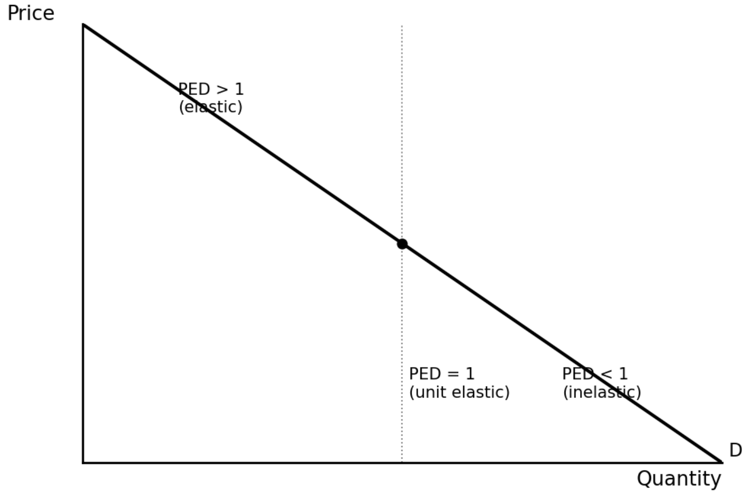
C The slope of the Lorenz curve at its midpoint.

D The maximum vertical distance from the line of equality.

Question 29 **[1 mark]**

Figure 10 shows a linear demand curve with elasticity regions marked.

Figure 10 — PED along a linear demand curve



Based on Figure 10, a profit-maximising monopolist would NEVER produce at a point where:

A Demand is inelastic, because a price rise would increase revenue and reduce costs.

B Demand is elastic.

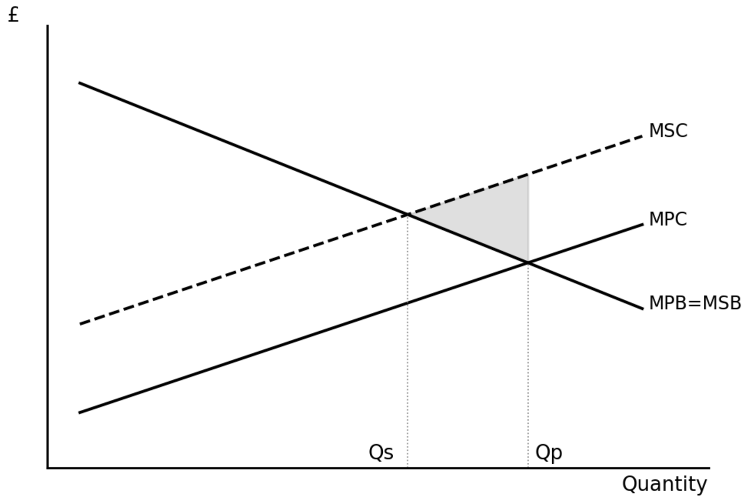
C PED = 1 (unit elastic).

D Quantity is at its maximum.

Question 30 **[1 mark]**

Figure 11 shows a negative production externality.

Figure 11 — Negative production externality



Based on Figure 11, a marketable emissions permit scheme set at the social optimum would:

A Leave output unchanged at Qp.

B Reduce output from Qp to Qs at the lowest-cost producers, achieving the efficient allocation.

C Raise output above Qp.

D Be economically equivalent to banning the activity entirely.

Section B

Answer all questions. Each question's marks are shown in brackets.

Study Extracts A, B, C and D, then answer Questions 31 to 33.

Case study: Artificial intelligence, labour markets and inequality

Extract A: The AI productivity boom

Figure 12: Selected indicators on AI and the UK labour market

Indicator	2020	2022	2024
UK business AI adoption rate (%)	12	24	43
Share of UK jobs 'highly exposed' to generative AI (%)	—	—	30
Real GDP per hour worked (2020 = 100)	100.0	101.5	102.8
Top 10% / bottom 10% earnings ratio	4.0	4.2	4.5
AI-related job vacancies (% of total)	1.1	2.8	5.4

Source: ONS / Bank of England / IFS, 2024

Extract B: AI, automation and labour demand

Generative artificial intelligence (AI) has advanced rapidly since 2022, with tools such as ChatGPT and specialised coding, writing and design assistants now used by over 40% of UK businesses. A 2024 IMF study estimated that 30% of UK jobs are 'highly exposed' to generative AI, with an additional 30% moderately exposed. Unlike earlier waves of automation that primarily displaced routine manual tasks, current AI technologies threaten cognitive work — including in accounting, law, journalism, translation and customer service. Labour economists disagree about the net effect: some studies suggest AI will complement (raise the productivity of) most workers, while others warn of substantial displacement.

Source: News reports, September 2024

Extract C: Inequality and the technology premium

Income inequality in the UK has risen since 2008, with the Gini coefficient drifting from around 0.34 to 0.36. A key driver has been the growing 'technology premium' — the gap between wages of workers with technical / AI-related skills and those without. In 2024, UK AI-related job vacancies offered average wages around 50% above the national median. Regional inequality has also widened: London and the South East capture a disproportionate share of AI-related investment and jobs, exacerbating the gap with lower-productivity regions.

The distributional pattern echoes earlier periods of technological change, when skill-biased technical change raised the relative demand for educated workers. However, some economists argue generative AI may differ because it can substitute for (rather than complement) cognitive skills. Acemoglu (2024) warns that if AI primarily substitutes for workers rather than empowers them, the labour share of national income may continue to fall. Government data shows UK business investment in computing and software has risen sharply since 2022, but investment in worker training has not kept pace.

Source: News reports, October 2024

Extract D: Policy options for the AI economy

Governments face difficult choices about how to respond to AI. One school argues for a non-interventionist stance: productivity gains from AI will raise average incomes, and markets will adjust through reallocation of labour to new sectors. The historical record of technology transitions — from the printing press to computing — shows that new technologies ultimately create more jobs than they destroy, even when short-term disruption is severe. In this view, supply-side reforms (skills training, labour market flexibility) are the main policy response needed.

Others advocate more active intervention. Proposed policies include: heavier regulation of AI developers to reduce concentration of market power; strengthening labour rights and collective bargaining so workers can capture more of the productivity gains; greater investment in retraining and income support for displaced workers; a universal basic income (UBI) funded by higher taxation of AI-related profits; or targeted taxation of capital substituting for labour ("robot tax"). Critics of these measures point to risks of government failure (robot tax would slow AI adoption, damaging UK competitiveness), regulatory capture (big AI firms lobbying to entrench their position), and unintended consequences (UBI may reduce labour supply and growth). A 2024 OECD report concluded that no single policy is sufficient — a combination of education reform, targeted transfers and competition policy is needed.

Source: News reports, December 2024

3 1 Using the data in the Extracts, examine the extent to which the rapid adoption of AI technologies has affected UK productivity and income inequality since 2020. **[10 marks]**

3 2 Using Extracts A, B and C and your knowledge of economics, explain why the adoption of AI technologies may increase both productivity and income inequality in the UK economy. **[15 marks]**

3 3 Considering the information in Extract D and the original evidence in Extracts A, B and C, do you recommend that the UK government should adopt active interventionist policies (such as labour-protective regulation, a universal basic income or targeted AI taxation) rather than relying on market adjustment to manage the economic effects of AI? Justify your recommendation. **[25 marks]**

END OF QUESTIONS